

CHAPTER - 3

PROBABILITY DISTRIBUTIONS

[Binomial, Poisson, Normal Distributions]

3.1 INTRODUCTION

We have already learnt frequency distributions which are based on the actual observations or experiment. In this chapter, we shall discuss theoretical distributions in which variates are distributed as per some definite law which can be expressed mathematically. That is, theoretical distributions (or Probability distributions) are mathematical models. So, it is possible to deduce mathematically what the frequency distribution of certain population should be. Such distributions are called Theoretical distributions or Frequency Distributions. Hence a theoretical distribution is the frequency distribution of a certain event in which frequencies are obtained by mathematical computation.

The value of the random variables and the corresponding probabilities are arranged such a way that one can suit a mathematical function for the probabilities in terms of the value of the random variable. A good model requires sufficiently large number of observations.

Just, as in the case of actual frequency distribution, the various statistical measures such as mean, standard deviation etc., are computed, so also such statistical measures may easily be calculated from a theoretical distribution and they give a very close approximation to such statistical measures as calculated from the actual frequency distribution.

There are two types of theoretical distributions (or Probability distributions)

(i) Discrete Theoretical Distributions

- (a) Binomial distribution
- (b) Poisson distribution
- (c) Rectangular distribution
- (d) Negative Binomial distribution
- (e) Geometric distribution

(ii) Continuous Theoretical Distributions

- (a) Normal distribution
- (b) Student's t-distribution
- (c) Chi-Square distribution
- (d) F-distribution

In this chapter, we shall study only the three distributions namely, Binomial, Poisson and Normal distributions. Before we discuss these distributions, we first give a brief introduction of uniform and Bernoulli's distributions.

3.2 DISCRETE UNIFORM DISTRIBUTION

A random variable X has a discrete uniform distribution if and only if its probability distribution is given by

$$p(x) = \frac{1}{k}, \text{ for } x = x_1, x_2, \dots, x_k.$$

The random variable X is then called discrete uniform random variable.

For example, the distributions

x	0	1
$p(x)$	$\frac{1}{2}$	$\frac{1}{2}$

and

x	0	1	2	3
$p(x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

are discrete uniform distributions.

3.3 BERNOULLI'S DISTRIBUTION

A random variable X which takes two values 0 and 1 with probability q and p respectively i.e., $P(X=0) = q$ and $P(X=1) = p$, $q = 1 - p$ is called a *Bernoulli's discrete random variable* and is said to have a *Bernoulli's distribution*. The probability function of Bernoulli's distribution can be written as

$$p(x) = p^x q^{1-x} = p^x (1-p)^{1-x}, x = 0, 1.$$

Note : (i) Mean of the Bernoulli's discrete random variable X is

$$\mu = E(X) = \sum x_i \cdot p(x_i) = 0 \times q + 1 \times p = p$$

(ii) Variance of X is

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 = \sum x_i^2 \cdot p(x_i) - \mu^2 = (0^2 \times q + 1^2 \times p) - p^2 \\ &= p - p^2 = p(1-p) = pq \end{aligned}$$

Hence the standard deviation is $\sigma = \sqrt{pq}$.

Definition : Suppose, associated with a random trial there is an event called 'success' and the complementary event called 'failure'. Let the probability for success be p and probability for failure be q . Suppose the random trials are prepared n times under identical conditions. These are called Bernoullian trials.

Bernoulli's Theorem :

If the probability of the occurrence of an event (success) in a single trial is p , then the probability that it will occur exactly r times out of n independent Bernoullian trials is

$${}^nC_r p^r q^{n-r} \text{ where } p + q = 1.$$

Proof : In a series of n independent trials, the number of r successes and $(n-r)$ failures can happen in nC_r mutually exclusive and exhaustive ways, since that is equal to the number of combinations of n things taken r at a time. Now consider a particular set of r successes and $(n-r)$ failures. Let the probability of success in a single trial be p . Then the probability of its failure is $q (= 1-p)$ and that of success in all the r trials is p^r , by the theorem of compound probability. Also, by the same theorem, the probability of failure in all the $(n-r)$ trials is q^{n-r} . Hence by the multiplication theorem, the probability of getting

r successes and $(n - r)$ failures in the particular set considered is equal to $p^r q^{n-r}$. Thus the probability of r successes in n trials with probability p in each trial is ${}^nC_r p^r q^{n-r}$.

Note : 1. By Binomial Theorem,

$$(q + p)^n = q^n + {}^nC_1 q^{n-1} p + {}^nC_2 q^{n-2} p^2 + \dots + {}^nC_r q^{n-r} p^r + \dots + p^n.$$

We notice that the probabilities of getting 0, 1, 2, ..., n successes are given respectively by the successive terms of the above binomial expansion. The probability of exactly r successes and $n - r$ failures is ${}^nC_r p^r q^{n-r}$ and it is the $(r + 1)^{\text{th}}$ term in the above binomial expansion.

2. The probability of getting at least one success in a series of n trials is

$$\begin{aligned} &= {}^nC_1 q^{n-1} p + {}^nC_2 q^{n-2} p^2 + \dots + p^n \\ &= (q + p)^n - q^n = 1 - q^n \end{aligned}$$

3. The probability of no success in n Bernoullian trials is given by q^n , and that of all successes by p^n .

BINOMIAL DISTRIBUTION

3.4 BINOMIAL DISTRIBUTION

Binomial distribution was discovered by James Bernoulli in the year 1700 and it is a discrete probability distribution.

Let us visualise a conceptual or practical situation where a trial or an experiment results in only two outcomes, say 'success' and 'Failure'. Further, the result of one trial does not influence the result of next trial, and the probability of success at each trial is the same from trial to trial.

Some of such situations are:

1. Tossing a coin - Head or Tail
2. Birth of a baby - Girl or Boy
3. Auditing a Bill - contains an error or not
4. An advertisement on TV - Recalled by viewer or not.

The conditions for the applicability of a Binomial distribution are as follows:

- (i) There are n independent trials
- (ii) Each trial has only two possible outcomes
- (iii) The probabilities of two outcomes remain constant.

The Binomial distribution can be described with the help of a function and is derived as follows:

Let the number of trials be n . The trials be independent i.e., the success or failure at one trial does not affect the outcome of the other trials. Thus the probability of success remains the same from trial to trial. Also let ' p ' be the probability of success and ' q ' be the probability of failure. Then we have $p + q = 1$ or $q = 1 - p$.

One could be interested in finding the probability of getting r successes, which implies getting $(n - r)$ failures.

One of the ways in which one can get r successes and $(n-r)$ failure is to get r continuous successes first and $(n-r)$ successive failures.

The probability of getting such a sequence of r successes and $(n-r)$ failures is,

$$= p^r q^{n-r} \text{ (using Multiplication Theorem of Probability)}$$

However, the number of ways in which one can get r successes and $n-r$ failures is nC_r and the probability for each of these ways is $p^r q^{n-r}$.

Thus, if x is the random variable representing the number of successes, the probability of getting r successes and $n-r$ failures, in n trials, is given by the probability function

$$P(x=r) = {}^nC_r p^r q^{n-r}, \quad r=0, 1, 2, \dots, n$$

This probability function is popularly known as the Binomial Distribution.

The probabilities of 0, 1, 2, ..., r , ..., n successes are therefore given by

$$q^n, {}^nC_1 p q^{n-1}, {}^nC_2 p^2 q^{n-2}, \dots, {}^nC_r p^r q^{n-r}, \dots, p^n.$$

The probability of the number of successes so obtained is called the *Binomial Probability Distribution*, because these probabilities are the successive terms in the expansion of the binomial $(q+p)^n$. This distribution contains two independent constants namely n and p (or q). They are called parameters of the Binomial Distribution. Sometimes, n is also known as the degree of the distribution.

Definition : A random variable X has a Binomial distribution if it assumes only non-negative values and its probability density function is given by

$$P(X=r) = p(r) = \begin{cases} {}^nC_r p^r q^{n-r}; & r=0, 1, 2, \dots, n; q=1-p \\ 0, & \text{otherwise} \end{cases}$$

Note :1. An alternate notation for binomial distribution is as follows : [JNTU (K) Dec. 2013 (Set No. 1)]

$$P(X=x) = p(x) = b(x; n, p) = \begin{cases} {}^nC_x p^x q^{n-x}; & x=0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

2. The Binomial distribution function is given by

$$F_X(x) = P(X \leq x) = \sum_{r=0}^x {}^nC_r p^r q^{n-r}$$

Examples of Binomial Distribution :

- (i) The number of defective bolts in a box containing ' n ' bolts.
- (ii) The number of machines lying idle in a factory having ' n ' machines.
- (iii) The number of post-graduates in a group of ' n ' men.
- (iv) The number of oil wells yielding natural gas in a group of ' n ' wells test drilled.

Conditions of Binomial Distribution :

The Binomial Distribution holds under the following conditions :

1. Trials are repeated under identical conditions for a fixed number of times, say n times.
2. There are only two possible outcomes, e.g. success or failure for each trial.
3. The probability of success in each trial remains constant and does not change from trial to trial.
4. The trials are independent i.e., the probability of an event in any trial is not affected by the results of any other trial.

3.5 CONSTANTS OF BINOMIAL DISTRIBUTION**1. Mean of the Binomial Distribution :**

[JNTU 2000, 2002, (K) Dec. 2013, Mar. 2014 (Set No. 4)]

The Binomial probability distribution is given by

$$p(r) = {}^nC_r p^r q^{n-r}, r = 0, 1, 2, \dots, n \text{ and } q = 1 - p$$

$$\text{Mean of } X, \mu = E(X) = \sum_{r=0}^n r p(r)$$

$$= 0 \times q^n + 1 \times {}^nC_1 p q^{n-1} + 2 \times {}^nC_2 p^2 q^{n-2} + \dots + n \cdot p^n$$

$$= np q^{n-1} + 2 \cdot \frac{n(n-1)}{2!} p^2 q^{n-2} + 3 \cdot \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} + \dots + np^n$$

$$= np \left[q^{n-1} + (n-1)p q^{n-2} + \frac{(n-1)(n-2)}{2!} p^2 q^{n-3} + \dots + p^{n-1} \right]$$

$$= np (q + p)^{n-1} \quad [\text{using Binomial Theorem}]$$

$$= np (1) = np \quad [\because p + q = 1]$$

Hence the Arithmetic mean of the Binomial distribution = np .

2. Variance of the Binomial Distribution :

[JNTU 2000, 2002, (K) Dec. 2013, Mar. 2014 (Set No. 4)]

$$\text{Variance, } V(X) = E(X^2) - [E(X)]^2$$

$$= \sum_{r=0}^n r^2 p(r) - \mu^2 = \sum_{r=0}^n [r(r-1) + r] \cdot p(r) - \mu^2$$

$$= \sum_{r=0}^n r(r-1) {}^nC_r p^r q^{n-r} + \sum_{r=0}^n r p(r) - \mu^2$$

$$= \left[2 \cdot {}^nC_2 p^2 q^{n-2} + 3 \cdot 2 \cdot {}^nC_3 p^3 q^{n-3} + \dots + n(n-1) p^n \right] + \mu - \mu^2$$

$$= \left[2 \cdot \frac{n(n-1)}{2!} p^2 q^{n-2} + 6 \cdot \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} + \dots + n(n-1) p^n \right] + \mu - \mu^2$$

$$= n(n-1)p^2 \left[q^{n-2} + (n-2)p q^{n-3} + \frac{(n-2)(n-3)}{2!} p^2 q^{n-4} + \dots + p^{n-2} \right] + \mu - \mu^2$$

$$\begin{aligned}
 \therefore V(X) &= n(n-1)p^2(q+p)^{n-2} + \mu - \mu^2 && \text{(by Binomial Theorem)} \\
 &= n(n-1)p^2(1)^{n-2} + \mu - \mu^2 && \text{(since } q+p=1) \\
 &= n(n-1)p^2 + np - (np)^2 && (\because \mu = np) \\
 &= n^2p^2 - np^2 + np - n^2p^2 = -np^2 + np = np(1-p) = npq
 \end{aligned}$$

$\therefore$ Variance of the Binomial Distribution $= npq$.

Hence the Standard Deviation of the Binomial Distribution $= \sqrt{npq}$.

3. Mode of the Binomial Distribution :

Mode of the binomial distribution is the value of x at which $p(x)$ has maximum value.

$$\text{Mode} = \begin{cases} \text{integral part of } (n+1)p, & \text{if } (n+1)p \text{ is not an integer} \\ (n+1)p \text{ and } (n+1)p-1, & \text{if } (n+1)p \text{ is an integer} \end{cases}$$

Recurrence Relation for the Binomial Distribution :

$$\text{We know that } P(r) = {}^nC_r p^r q^{n-r} \quad \dots (1)$$

$$\therefore P(r+1) = {}^nC_{r+1} p^{r+1} q^{n-r-1} \quad \dots (2)$$

(2) $\div$ (1) gives

$$\frac{P(r+1)}{P(r)} = \frac{{}^nC_{r+1}}{{}^nC_r} \cdot \frac{p^{r+1}q^{n-r-1}}{p^r q^{n-r}} = \frac{n-r}{r+1} \cdot \frac{p}{q}$$

$$\text{or } P(r+1) = \frac{(n-r)p}{(r+1)q} \cdot P(r)$$

3.6 BINOMIAL FREQUENCY DISTRIBUTION

If n independent trials constitute one experiment and this experiment is repeated N times, then the frequency of r successes is $N \cdot {}^nC_r p^r q^{n-r}$. Since the probabilities of 0, 1, 2, ..., r , ..., n successes in n trials are given by the terms of the binomial expansion of $(q+p)^n$, therefore in N sets of n trials the theoretical frequencies of 0, 1, 2, ..., r , ..., n successes will be given by the terms of expansion of $N(q+p)^n$. The possible number of successes and their frequencies is called a Binomial Frequency Distribution.

SOLVED EXAMPLES

Example 1 : A fair coin is tossed six times. Find the probability of getting four heads.

Solution : p = probability of getting a head $= 1/2$

q = probability of not getting head $= 1/2$

and $n = 6, r = 4$

We know that $P(r) = {}^nC_r p^r q^{n-r}$

$$\therefore P(4) = {}^6C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{6-4} = \frac{6!}{4!2!} \cdot \left(\frac{1}{2}\right)^6 = \frac{6 \times 5}{2} \cdot \frac{1}{2^6} = \frac{15}{64} = 0.2344.$$

Example 2 : Determine the probability of getting the sum 6 exactly 3 times in 7 throws with a pair of fair dice.

Solution : In a single throw of a pair of fair dice, a sum of 6 can occur in 5 ways : (1, 5), (5, 1), (2, 4), (4, 2) and (3, 3) out of $6 \times 6 = 36$ ways.

Thus p = Probability of occurrence of 6 in one throw = $\frac{5}{36}$

$$q = 1 - p = 1 - \frac{5}{36} = \frac{31}{36}$$

n = Number of trials = 7

$\therefore$ Probability of getting 6 exactly thrice in 7 throws

$$= {}^7C_3 p^3 q^{7-3} = {}^7C_3 \left(\frac{5}{36}\right)^3 \left(\frac{31}{36}\right)^4 = \frac{35(125)(31)^4}{(36)^7} = 0.0516 \text{ (nearly).}$$

Example 3 : A die is thrown 6 times. If getting an even number is a success, find the probabilities of (i) at least one success (ii) ≤ 3 successes (iii) 4 successes.

Solution : In a single throw of a die, an even number can occur in 3 ways out of 6 ways. Thus

p = Probability of occurrence of an even number in one throw = $\frac{3}{6} = \frac{1}{2}$

n = Number of trials = 6

(i) $P(r \geq 1) = 1 - P(r = 0)$

$$= 1 - \left(\frac{1}{2}\right)^6 = 1 - \frac{1}{64} = \frac{63}{64} = 0.9844$$

(ii) $P(r \leq 3) = P(r = 0) + P(r = 1) + P(r = 2) + P(r = 3)$

$$= \left(\frac{1}{2}\right)^6 ({}^6C_0 + {}^6C_1 + {}^6C_2 + {}^6C_3) = \frac{21}{32} = 0.6562$$

(iii) $P(r = 4) = {}^6C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 = \frac{15}{64} = 0.2344$

Example 4 : Ten coins are thrown simultaneously. Find the probability of getting at least (i) seven heads [JNTU 1999, 2007, 2008S, (K) May 2013 (Set No. 2)]

(ii) six heads [JNTU (A) Dec, 2009, (K) May 2013 (Set No. 2)]

Solution : p = Probability of getting a head = $1/2$

q = Probability of not getting a head = $1/2$

The probability of getting r heads in a throw of 10 coins is

$$P(X=r) = p(r) = {}^{10}C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{10-r} ; r = 0, 1, 2, \dots, 10$$

(i) Probability of getting at least seven heads is given by $P(X \geq 7)$

$$= P(X=7) + P(X=8) + P(X=9) + P(X=10)$$

$$\begin{aligned}
&= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{10-7} + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8} + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \\
&= \frac{1}{2^{10}} [{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}] = \frac{1}{2^{10}} (120 + 45 + 10 + 1) \\
&= \frac{176}{1024} = 0.1719
\end{aligned}$$

(ii) This is left as an exercise to the student.

Example 5 : Two dice are thrown five times. Find the probability of getting 7 as sum

i) at least once ii) Two times iii) $P(1 < X < 5)$. [JNTU (H) Nov. 2009 (Set No. 2)]

Solution : p = The probability of getting a sum 7 in a single throw of a pair of dice

$$= \frac{6}{36} = \frac{1}{6}$$

q = The probability of not getting a sum 7 in a throw of a pair of dice

$$= 1 - p = 1 - \frac{1}{6} = \frac{5}{6}$$

Number of trials, $n = 5$.

(i) $P(\text{at least once}) = P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - {}^5C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^5 = 1 - \left(\frac{5}{6}\right)^5$$

$$\begin{aligned}
\text{(ii) } P(\text{exactly two times}) &= P(X = 2) = {}^5C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 \\
&= 10 \cdot \frac{5^4}{6^6} = \frac{2}{6} \cdot \left(\frac{5}{6}\right)^5 = \frac{1}{3} \cdot \left(\frac{5}{6}\right)^5
\end{aligned}$$

(iii) $P(1 < X < 5) = P(X = 2) + P(X = 3) + P(X = 4)$

$$\begin{aligned}
&= {}^5C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 + {}^5C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^2 + {}^5C_4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^1 \\
&= 10 \cdot \frac{5^4}{6^6} + 10 \cdot \frac{5^3}{6^6} + 5 \cdot \frac{5^2}{6^6} = \frac{5^3}{6^6} (300 + 100 + 50) = \frac{5^3}{6^6} (450) = 311 \left(\frac{5^3}{6^6}\right)
\end{aligned}$$

Example 6 : It has been claimed that in 60% of all solar heat installations the utility bill is reduced by at least one-third. Accordingly, what are the probabilities that the utility bill will be reduced by at least one-third in (i) four of five installations (ii) at least four of five installations.

[JNTU 1999S, (A) Apr. 2012 (Set No. 4)]

Solution : p = The probability that in the solar heat installations the utility bill is reduced by one-third
 $= 60\% = 0.6$

$$q = 1 - p = 1 - 0.6 = 0.4$$

(i) $r = 4, n = 5$

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

$$\therefore P(X=4) = {}^5C_4(0.6)^4(0.4)^{5-4} = 5(0.6)^4(0.4) = 0.2592$$

$$\begin{aligned} \text{(ii) } P(X \geq 4) &= P(X=4) + P(X=5) \\ &= 0.2592 + {}^5C_5(0.6)^5 \quad [\text{From (i)}] \\ &= 0.2592 + 0.07776 = 0.337 \end{aligned}$$

Example 7 : If 3 of 20 tyres are defective and 4 of them are randomly chosen for inspection, what is the probability that only one of the defective tyre will be included ?

[JNTU 2000S]

Solution : Let p = Probability of a defective tyre (success) = $3/20$

Then q = Probability of a non-defective tyre

$$= 1 - p = 1 - \frac{3}{20} = \frac{17}{20}$$

Given $n = 4$

The probability that exactly one tyre will be defective

$$= P(r=1) = {}^4C_1 \left(\frac{3}{20} \right)^1 \left(\frac{17}{20} \right)^{4-1} = \frac{4 \cdot 3 \cdot 17^3}{(20)^4} = 0.3685$$

Example 8 : The incidence of an occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of 6 workers chosen at random, four or more will suffer from the disease ?

[JNTU 2001 S, (K) Nov. 2011 (Set No. 4)]

Solution : The probability of a worker suffering from disease, $p = 20\% = 0.2$

$\therefore$ The probability that no worker suffering from disease, $q = 1 - p = 0.8$

No. of workers, $n = 6$

The probability that four or more workers will suffer from disease

$$\begin{aligned} &= P(X \geq 4) \\ &= P(X=4) + P(X=5) + P(X=6) \\ &= {}^6C_4(0.2)^4(0.8)^2 + {}^6C_5(0.2)^5(0.8) + {}^6C_6(0.2)^6 \\ &= (0.2)^4 [15(0.8)^2 + 6(0.2)(0.8) + (0.2)^2] \\ &= (0.2)^4 [9.6 + 0.96 + 0.4] = 0.0175 \end{aligned}$$

Example 9 : Determine the binomial distribution for which the mean is 4 and variance 3

[JNTU 1999]

Solution : Given mean of the distribution = 4 i.e., $np = 4$... (1)

and variance of the distribution = 3 i.e., $npq = 3$... (2)

$$(2) \div (1) \text{ gives } q = 3/4$$

$$\therefore p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$$

From (1), $n = \frac{4}{p} = 16$

$\therefore$ The given Binomial Distribution has parameters $n = 16$ and $p = 1/4$.

Note : To find mode of the distribution, we have $(n + 1)p = \frac{17}{4} = 4.25$, which is not an integer.

Hence the unique mode of the distribution = integral part of $(n + 1)p$ i.e., 4.25 = 4

Example 10 : If the probability of a defective bolt is $1/8$, find :

(i) The mean (ii) The variance for the distribution of defective bolts of 640.

[JNTU (H) Nov. 2009 (Set No. 4)]

Solution : We are given

p = The probability of a defective bolt = $1/8$ and $n = 640$.

$\therefore$ Mean of the distribution, $\mu = np = \frac{640}{8} = 80$

Also $q = 1 - p = 1 - \frac{1}{8} = \frac{7}{8}$.

Hence variance of the distribution = $npq = (np)q = \mu q = 80 \times \frac{7}{8} = 70$

Example 11 : In 256 sets of 12 tosses of a coin, in how many cases one can expect heads and 4 tails.

[JNTU 2003, 2004 (Set No. 4)]

Solution : The probability of getting a head, $p = 1/2$

$\therefore$ The probability of getting a tail, $q = 1/2$

Here $n = 12$

The probability of getting 8 heads and 4 tails in 12 trails

$$= P(X = 8) = {}^{12}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^4$$

$$= \frac{12!}{8!4!} \cdot \left(\frac{1}{2}\right)^{12} = \frac{495}{(2)^{12}}$$

$\therefore$ The expected number of such cases in 256 sets

$$= 256 \times P(X = 8) = 2^8 \times \frac{495}{2^{12}} = \frac{495}{16} = 30.9375 \sim 31$$

Example 12 : If the ...

Example 36 : The probability that John hits a target is $\frac{1}{2}$. He fires 6 times. Find the probability that he hits the target (i) exactly 2 times (ii) more than 4 times (iii) at least once.

[JNTU 2007S, (H) May 2011 (Set No. 4)]

Solution : Probability of hitting a target = $p = \frac{1}{2}$

Probability of no hit (or failure) = $q = \frac{1}{2}$

Number of trials = $n = 6$

Number of hits (successes) = X

(i) $P(\text{exactly 2 times}) = P(X = 2)$

$$= {}^6C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^4 = \frac{15}{2^6} = 0.2344$$

(ii) $P(\text{more than 4 times}) = P(X > 4) = P(X = 5) + P(X = 6)$

$$= {}^6C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right) + {}^6C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^0$$

$$= \frac{6}{2^6} + \frac{1}{2^6} = \frac{7}{2^6} = 0.1094$$

(iii) $P(\text{at least once}) = P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - {}^6C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^6 = 1 - \frac{1}{2^6} = 0.9844$$

Example 37 : The mean of Binomial distribution is 3 and the variance is $\frac{9}{4}$. Find

(i) the value of n (ii) $P(X \geq 7)$ (iii) $P(1 \leq X < 6)$.

[JNTU 2007, 2008S, (H) Dec. 2009 (Set No. 2)]

Solution : We have

Mean of the Binomial distribution = 3 $\Rightarrow np = 3$... (1)

Variance of the Binomial distribution = $\frac{9}{4} \Rightarrow npq = \frac{9}{4}$... (2)

(2) $\div$ (1) gives

$$\frac{npq}{np} = \frac{9}{4 \times 3} \Rightarrow q = \frac{3}{4}$$

$$\therefore p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\text{From (1), } n = \frac{3}{p} = 12$$

$$\begin{aligned}
 \text{(ii) } P(X \geq 7) &= 1 - P(X < 7) \\
 &= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3) \\
 &\quad + P(X=4) + P(X=5) + P(X=6)] \\
 &= 1 - \left[{}^{12}C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{12} + {}^{12}C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{11} + {}^{12}C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{10} \right. \\
 &\quad \left. + {}^{12}C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^9 + {}^{12}C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^8 + {}^{12}C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7 + {}^{12}C_6 \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^6 \right] \\
 &= 1 - \left[\left(\frac{3}{4}\right)^{12} + 12 \cdot \frac{1}{4} \left(\frac{3}{4}\right)^{11} + 66 \cdot \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{10} + 220 \cdot \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^9 \right. \\
 &\quad \left. + 495 \cdot \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^8 + 72 \cdot \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7 + 84 \cdot \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^6 \right] \\
 &= 1 - \left[\frac{3^7}{4^{12}} (243 + 972 + 1782 + 1980 + 1485 + 72 + 28) \right] \\
 &= 1 - 0.8554 = 0.1446
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) } P(1 \leq X < 6) &= P(X=1) + \dots + P(X=5) \\
 &= {}^{12}C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^{11} + {}^{12}C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{10} + {}^{12}C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^9 \\
 &\quad + {}^{12}C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^8 + {}^{12}C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7 \\
 &= \frac{3^7}{4^{12}} (972 + 1782 + 1980 + 1485 + 72) \\
 &= \frac{3^7}{4^{12}} (6291) = 0.82
 \end{aligned}$$

Example 38 : The probability that the life of a bulb is 100 days is 0.05. Find the probability that out of 6 bulbs (i) At least one (ii) greater than four (iii) none, will be having a life of 100 days.

[JNTU 2007S, (K) May 2013 (Set No. 3)]

Solution : We have

$$p = \text{Probability that a bulb is of 100 days life} = 0.05 = \frac{1}{20} \text{ and } n = 6$$

$$\therefore q = 1 - p = 1 - \frac{1}{20} = \frac{19}{20}$$

By Binomial distribution,

$$P(X=x) = {}^nC_x p^x q^{n-x} = {}^6C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{6-x}$$

$$(i) P(x \geq 1) = 1 - P(x=0) = 1 - {}^6C_0 \left(\frac{1}{20}\right)^0 \left(\frac{19}{20}\right)^6 = 1 - \left(\frac{19}{20}\right)^6 = 0.265$$

$$\begin{aligned} (ii) P(x > 4) &= P(x=5) + P(x=6) \\ &= {}^6C_5 \left(\frac{1}{20}\right)^5 \cdot \frac{19}{20} + {}^6C_6 \left(\frac{1}{20}\right)^6 \\ &= \left(\frac{1}{20}\right)^6 [114 + 1] = \frac{115}{(20)^6} \end{aligned}$$

$$(iii) P(x=0) = {}^6C_0 \left(\frac{1}{20}\right)^0 \left(\frac{19}{20}\right)^6 = \left(\frac{19}{20}\right)^6 = 0.7351$$

Example 39 : If the chance that any of the 10 telephone lines is busy at an instant is 0.2. What is the most probable number of busy lines and what is the probability of this number.

[JNTU 2008 (Set No. 4)]

Solution : Given the probability that a line is busy is $p = 0.2$

$\therefore$ The probability that a line is not busy is $q = 1 - p = 1 - 0.2 = 0.8$

On average $np = 10(0.2) = 5$ lines are busy at an instant.

The probability that ' r ' lines are busy is

$$P(r) = {}^{10}C_r (0.2)^r (0.8)^{10-r}$$

$$\begin{aligned} \therefore P(5 \text{ lines are busy}) &= P(5) = {}^{10}C_5 (0.2)^5 (0.8)^{10-5} \\ &= {}^{10}C_5 (0.2)^5 (0.8)^5 \end{aligned}$$

Example 40 : Seven coins are tossed and the number of heads are noted. The experiment is repeated 128 times and the following distribution is obtained.

No. of heads	0	1	2	3	4	5	6	7	Total
Frequency	7	6	19	35	30	23	7	1	128

Fit a binomial distribution assuming. (a) the coin is unbiased.

(b) the nature of the coin is not known.

[JNTU 2009 (Set No. 1)]

Solution : (a) The coin is unbiased.

$$\therefore p = \frac{1}{2}, q = \frac{1}{2} \text{ and } n = 7$$

$$N = \sum f_i = 7 + 6 + 19 + 35 + 30 + 23 + 7 = 128$$

By binomial distribution, $p(x) = {}^nC_x p^x q^{n-x}$

We have the recurrence relation

$$p(x+1) = \frac{(n-x)p}{(x+1)q} \cdot p(x) = \frac{7-x}{x+1} \cdot p(x) \left[\because n=7, \frac{p}{q}=1 \right]$$

$$\therefore p(0) = {}^7C_0 p^0 q^7 = {}^7C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^7 = \frac{1}{2^7}$$

No. of heads x	Observed frequency	Probability $p(x)$	Expected or Theoretical frequency $f(x) = N \cdot p(x)$
0	7	$p(0) = \frac{1}{2^7}$	$f(0) = 128 \cdot p(0) = 128 \times \frac{1}{2^7} = 1$
1	6	$p(1) = 7 \cdot p(0) = \frac{7}{2^7}$	$f(1) = 128 \cdot p(1) = 128 \times \frac{7}{2^7} = 7$
2	10	$p(2) = 3 \cdot p(1) = \frac{21}{2^7}$	$f(2) = 128 \cdot p(2) = 128 \times \frac{21}{2^7} = 21$
3	35	$p(3) = \frac{35}{2^7}$	$f(3) = 128 \cdot p(3) = 35$
4	30	$p(4) = \frac{35}{2^7}$	$f(4) = 128 \cdot p(4) = 35$
5	23	$p(5) = \frac{21}{2^7}$	$f(5) = 128 \cdot p(5) = 21$
6	7	$p(6) = \frac{7}{2^7}$	$f(6) = 128 \cdot p(6) = 7$
7	1	$p(7) = \frac{1}{2^7}$	$f(7) = 128 \cdot p(7) = 1$

(b) The nature of the coin is not known.

We are given $n = 7$ and $N = \sum f_i = 128$

$$\therefore \text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{6 + 38 + 105 + 120 + 115 + 42 + 7}{128}$$

$$= \frac{433}{128} = 3.383$$

i.e., $np = 3.383$

$$\Rightarrow 7p = 3.383 \quad \therefore p = \frac{3.383}{7} = 0.4833$$

and $q = 1 - p = 1 - 0.4833 = 0.5167$

Hence the Binomial Distribution to be fitted is given by the terms of

$$N(q + p)^n = 128 (0.5167 + 0.4833)^7$$

This information can also be tabulated as in the earlier table.

Example 41: 30% of items from a factory are defective. Find the probability that in a sample of 8 (i) one (ii) At least two (iii) $p(1 < x < 6)$ are defective.

[JNTU (H) 2010 (Set No. 1)]

Solution: Given probability of a defective item = $30\% = \frac{30}{100} = \frac{3}{10} = 0.3$

$$\therefore q = 1 - p = 1 - 0.3 = 0.7$$

n = Sample size = 8.

For a Binomial Distribution, the probability function is

$$P(X=r) = f(x) = {}^nC_r p^r q^{n-r} = {}^8C_r (0.3)^r (0.7)^{8-r}$$

$$(i) P(X=1) = {}^8C_1 (0.3) (0.7)^{8-1} = 8(0.3) (0.7)^7 = 0.1976$$

$$(ii) P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X=0) + P(X=1)]$$

$$= 1 - [{}^8C_0 (0.3) (0.7)^8 + 0.1976] = 1 - [(0.3) (0.7)^8 + 0.1976]$$

$$= 1 - [0.0173 + 0.1976] = 1 - 0.3706 = 0.6294.$$

$$(iii) P(1 < X < 6) = P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= {}^8C_2 (0.3)^2 (0.7)^{8-2} + {}^8C_3 (0.3)^3 (0.7)^{8-3} + {}^8C_4 (0.3)^4 (0.7)^{8-4} + {}^8C_5 (0.3)^5 (0.7)^{8-5}$$

$$= \frac{8!}{2!6!} (0.3)^2 (0.7)^6 + \frac{8!}{3!5!} (0.3)^3 (0.7)^5 + \frac{8!}{4!4!} (0.3)^4 (0.7)^4 + \frac{8!}{5!3!} (0.3)^5 (0.7)^3$$

$$= 0.0267 + 0.2541 + 0.1361 + 0.0467 = 0.4636.$$

Example 42: The probability that a man hitting a target is $1/3$. If he fires 6 times, find the probability that he fires (i) At the most 5 times (ii) Exactly once (iii) At least two times

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution: Given p = Probability of hitting the target = $\frac{1}{3}$ and $n = 6$.

$$\therefore q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

For a Binomial Distribution, the probability function is

$$f(x) = P(X=x) = {}^nC_x p^x q^{n-x} = {}^6C_x p^x q^{6-x}$$

$$(i) P(x \leq 5) = 1 - P(x=6) = 1 - {}^6C_6 \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^{6-6} = 1 - \frac{1}{729} = \frac{728}{729} = 0.9986$$

$$(ii) P(x=1) = {}^6C_1 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^{6-1} = 2 \left(\frac{2}{3}\right)^5 = 0.2634$$

$$(iii) P(x \geq 2) = 1 - [P(x=0) + P(x=1)] = 1 - \left[{}^6C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{6-0} + 0.2634 \right] \text{ [from (i)]}$$

$$= 1 - \left[6 \left(\frac{2}{3}\right)^6 + 0.2634 \right] = 1 - [0.5267 + 0.2634] = 1 - 0.7901 = 0.2099.$$

Example 43 : A coin is biased in a way that a head is twice as likely to occur as a tail. If the coin is tossed 3 times, find the probability of getting 2 tail and 1 head

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution : When a coin is tossed, two mutually exclusive events H, T can occur.

We have $P(H) + P(T) = 1$

$$P(H) = 2 P(T) \Rightarrow 3 P(T) = 1$$

$$\Rightarrow P(T) = \frac{1}{3} = p \quad (\text{say probability of success})$$

$$\Rightarrow P(H) = \frac{2}{3} = q \quad (\text{say probability of failure})$$

In a Binomial distribution $n = 3$, Number of successes = $x = 2$.

$\therefore$ Required probability is

$$P(X = x) = {}^nC_x p^x q^{n-x} = {}^3C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^1 = 3 \times \frac{1}{9} \times \frac{2}{3} = \frac{2}{9}$$

Example 44 : Traffic control engineer reports that 75% of the vehicles passing through a checkpoint are from within state. What is the probability that fewer than 4 of the 9 are from out of the state?

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : The probability that vehicle is from within state = $\frac{75}{100} = \frac{3}{4}$

Considering as binomial distribution, $n = 9$

The probability more than three vehicles are from within state = $P(X \geq 3) = 1 - P(X < 3)$

$$= 1 - P(X = 0) - P(X = 1) - P(X = 2)$$

$$= 1 - {}^9C_0 \left(\frac{3}{4}\right)^0 \left(\frac{1}{4}\right)^9 - {}^9C_1 \left(\frac{3}{4}\right)^1 \left(\frac{1}{4}\right)^8 - {}^9C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^7$$

Example 45 : Find the probability of getting an even number 3 or 4 or 5 times in throwing 10 dice. using Binomial Distribution.

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let p be the probability of getting an even number in a throw of a die.

$$\text{Then } p = \frac{3}{6} = \frac{1}{2} \text{ and } q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$$

Here $n = 10$

The probability of getting x even numbers in ten throws of a die is

$$P(X=x) = {}^{10}C_x p^x q^{10-x}, x=0,1,2,\dots,10$$

$$= {}^{10}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x} = {}^{10}C_x \left(\frac{1}{2}\right)^{10}; x=0,1,2,\dots,10$$

$$\therefore P(X=3) = {}^{10}C_3 \left(\frac{1}{2}\right)^{10} = \frac{120}{1024} = 0.112$$

$$P(X=4) = {}^{10}C_4 \left(\frac{1}{2}\right)^{10} = \frac{210}{1024} = 0.2$$

$$P(X=5) = {}^{10}C_5 \left(\frac{1}{2}\right)^{10} = \frac{252}{1024} = 0.246$$

Example 46 : A die is thrown 8 times. If getting a 2 or 4 is a success. Find the probability of (i) 4 success (ii) $P(x \leq 3)$ (iii) $P(x \geq 2)$ [JNTU (A) Nov. 2011 (Set No. 3)]

Solution : We have

p = The probability of obtaining a 2 or 4 in one throw (success) $= \frac{2}{6} = \frac{1}{3}$

$$\therefore q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

n = Number of trials = 8.

The probability of x successes is given by

$$P(X=x) = p(x) = {}^8C_x p^x q^{8-x}$$

$$(i) \quad p(x=4) = {}^8C_4 p^4 q^{8-4} = {}^8C_4 p^4 q^4$$

$$= \frac{8!}{4!4!} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^4 = 0.17$$

$$(ii) \quad p(x \leq 3) = p(x=0) + p(x=1) + p(x=2) + p(x=3)$$

$$= q^8 + 8pq^7 + {}^8C_2 p^2 q^6 + {}^8C_3 p^3 q^5$$

$$= \left(\frac{2}{3}\right)^8 + 8\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^7 + 28\left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^6 + 56\left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^5$$

$$= \frac{1}{3^8} [2^8 + 8(2)^7 + 28(2^6) + 56(2^5)]$$

$$= \frac{2^5}{3^8} [8 + 32 + 56 + 56] = \frac{32}{6561} (152) = 0.74$$

$$(iii) \quad p(x \geq 2) = 1 - p(x < 2)$$

$$= 1 - [p(x=0) + p(x=1)] = 1 - [q^8 + 8pq^7]$$

$$= 1 - \left[\left(\frac{2}{3}\right)^8 + 8\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^7 \right] = 1 - \frac{1}{3^8} [2^8 + 8 \cdot 2^7]$$

$$= 1 - \frac{2^8}{3^8} (1 + 4) = 1 - \left(\frac{2}{3}\right)^8 (5) = 1 - 0.1951 = 0.8049$$

Example 47 : In a family of 5 children find the probability that there are

- (i) 2 boys (ii) At least one boy (iii) All are boys (iv) No boys.

[JNTU(A) Nov. 2011 (Set No. 4)]

Solution : Here n = number of children = 5

p = The probability of getting a boy = $\frac{1}{2}$ [$\because$ Equal probability for boy and girl]

$$\therefore q = 1 - p = \frac{1}{2}$$

The probability $P(x)$ of x successes in a series of n independent trials is given by

$$P(x) = {}^nC_x p^x q^{n-x}$$

$$\therefore P(x) = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

$$(i) \quad P(2 \text{ boys}) = p(x=2) = {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2} = \frac{10}{2^5} = \frac{5}{16}$$

$$(ii) \quad P(\text{at least one boy}) = p(x \geq 1) = 1 - p(x=0)$$

$$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0} = 1 - \frac{1}{32} = \frac{31}{32}$$

$$(iii) \quad P(\text{all are boys}) = p(x=5) = {}^5C_5 \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

$$(iv) \quad P(\text{no boy}) = p(x=0) = {}^5C_0 \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Example 48 : A box contains 9 cards numbered 1 to 9. If four cards are drawn with replacement. What is the probability that none is 1 ?

[JNTU(A) Apr 2012 (Set No. 2)]

Solution : Given $n = 4$

Let $p =$ The probability of getting '1' $= \frac{1}{9}$

$$\text{Then } q = 1 - p = 1 - \frac{1}{9} = \frac{8}{9}$$

The probability p of x successes is given by $P(X = x) = {}^nC_x p^x q^{n-x}$

Hence the probability of getting none is 1 is given by

$$P(X = 0) = {}^4C_0 \left(\frac{1}{9}\right)^0 \left(\frac{8}{9}\right)^4 = \left(\frac{8}{9}\right)^4 = 0.624$$

Example 49 : It has been claimed that in 60% of all solar heat installations the utility bill is reduced by at least one third. Accordingly what are the probabilities that the utility bill will be reduced by at least one - third in (i) four of five installations. (ii) at least four of five installations.

[JNTU(A) Apr 2012 (Set No. 4)]

Solution : Let $p =$ The probability that in the solar heat installations the utility bill is reduced by one - third.

$$= 60\% = 0.6$$

$$\text{Then } q = 1 - p = 1 - 0.6 = 0.4$$

- (i) The probability that the utility bill will be reduced by atleast one - third in four of five installations is given by

$$\begin{aligned} P(X = 4) &= {}^5C_4 (0.6)^4 (0.4)^{5-4} \quad [\because P(X = r) = {}^nC_r p^r q^{n-r}] \\ &= 5(0.6)^4 (0.4) = 2(0.6)^4 = 0.2592 \end{aligned}$$

- (ii) The probability that the utility bill will be reduced by atleast one-third in atleast four of five installations is given by

$$\begin{aligned} P(X \geq 4) &= P(X = 4) + P(X = 5) \\ &= 0.2592 + {}^5C_5 (0.6)^5 (0.4)^{5-5}, \text{ using (i)} \\ &= 0.2592 + (0.6)^5 = 0.2592 + 0.07776 = 0.33696 \end{aligned}$$

Example 50 : The probability of a man hitting a target is $1/4$. If he fixes 7 times, the probability of hitting the target at least twice is _____

[JNTU (A) Dec. 2014]

Solution : The probability of a man hitting a target is $p = \frac{1}{4}$

$$\therefore q = 1 - p = 1 - \frac{1}{4} = \frac{3}{4}$$

Also we are given $n = 7$

$\therefore$ Required probability $= P(X \geq 2) = 1 - P(X < 2)$

$$= 1 - [P(X = 0) + P(X = 1)] = 1 - {}^nC_0 p^0 q^7 - {}^nC_1 p q^6$$

$$= 1 - {}^7C_0 \left(\frac{3}{4}\right)^7 - {}^7C_1 \frac{1}{4} \left(\frac{3}{4}\right)^6 = 1 - \left(\frac{3}{4}\right)^7 - \frac{7}{4} \left(\frac{3}{4}\right)^6$$

$$= 1 - \frac{3^6}{4^7} [3 + 7] = 1 - \frac{7290}{16384} = \frac{4547}{8192}$$

REVIEW QUESTIONS

1. Define Binomial Distribution.
2. Show that the mean of the Binomial distribution is the product of the parameter p and the number of times n .
3. What is the formula of mean and variance of a Binomial distribution.
4. Find the mean and variance of the Binomial distribution.
5. Prove that the variance of the Binomial distribution is npq .

EXERCISE 7(A)

1. A box contains 9 cards numbered 1 to 9. If four cards are drawn with replacement, find the probability that none is 1. [JNTU (A) Apr. 2012 (Set No. 1)]
2. In tossing ten coins, what is the probability of having exactly 5 heads?
3. A die is thrown five times. If getting an odd number is a success, find the probability of getting at least four successes.
4. The mean and variance of a binomial distribution are 4 and $4/3$ respectively. Find $P(X \geq 2)$.
5. For a binomial distribution, prove that the variance is less than the mean.
6. The mean and variance of a binomial distribution are 6 and 3 respectively. Find the mode of the binomial distribution.
7. 10% of screws produced in a factory turn out to be defective. Find the probability that in a sample of 10 screws chosen at random, exactly two will be defective.
8. If 10% of bolts produced by a machine are defective, find the probability that out of 10 bolts chosen at random (i) none will be defective (ii) one will be defective, and (iii) at most two bolts will be defective.
9. Eight fair coins are thrown simultaneously. Find the probability of getting (i) exactly six heads (ii) at most six heads.
10. Find the probability of (i) at least 2 heads (ii) at most 4 heads, in a single toss of 6 fair coins.
11. A die is thrown 8 times. Find the probability that 3 will appear (i) exactly 2 times (ii) at least seven times (iii) at least once.

12. The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that (i) exactly two will be defective (ii) at least two will be defective (iii) none will be defective.
- [JNTU (K) Nov. 2012 (Set No. 3)]
13. If the chance that one of the ten telephone lines is busy at an instant is 0.2,
 (i) What is the chance that 5 of the lines are busy ?
 (ii) What is the most probable number of busy lines and what is the probability of this number ?
 (iii) What is the probability that all the lines are busy ?
14. A box contains 100 transistors, 20 of which are defective and 10 are selected at random, find the probability that
 (i) all are defective (ii) at least one is defective
 (iii) all are good (iv) at most 3 are defective.
15. In a certain city only 50% of the students are capable of doing college work actually go to college. Assuming that this claim is true, find the probability that among 18 such a capable students
 (i) exactly 10 will go to college (ii) at least 2 will go to college
 (iii) at most 17 will go to college.
16. The probability of a newly generated virus attacked to the computer system will corrupt the 4 files out of 20 files which are opened in an hour. If 20 files are opened in an hour, find the probability that
 (i) at least 10 files are corrupted (ii) exactly 3 files are corrupted
 (iii) all files are corrupted (iv) all files are safe
 (v) more than 2 but not more than 5 files are corrupted.
17. The overall percentage of failures in a certain examination is 20. If six candidates appear in the examination, what is the probability that at least five pass the examination ?
18. A pair of dice is thrown. Find the probability of getting a sum 11 (i) once (ii) twice.
19. Find the probability that in tossing a fair coin 6 times there will appear (i) 0 (ii) 1 (iii) 2 (iv) 3 (v) 4 (vi) 5 and (vii) 6 heads.
20. 5 coins are tossed at a time, 3200 times. Find the frequencies of the distribution of heads and tails and tabulate the results. Also determine the mean number of success and standard deviation.
21. Fit a Binomial Distribution to the following data

x	0	1	2	3	4	5
f	38	144	342	287	164	25

[JNTU 2004S (Set No.2)]

22. The following data show the number of seeds germinating out of 10 on damp filter for 80 set of seeds. Fit a Binomial Distribution to the data.

x	0	1	2	3	4	5	6	7	8	9	10
f	6	20	28	12	8	6	0	0	0	0	0

[JNTU (K) May 2012 (Set No. 4)]

23. 7 coins are tossed at a time, 128 times. The number of heads observed at each throw is recorded and the results are given below :

No. of heads	0	1	2	3	4	5	6	7
Frequency	7	6	19	35	30	23	7	1

Fit a binomial distribution to the data assuming that the coins are (i) biased (ii) unbiased ?

[JNTU 2000 S]

24. Fit a Binomial Distribution to the following frequency data.

x	0	1	2	3	4
f	28	62	46	10	4

[JNTU Jan. 2007]

25. In 800 families with 5 children each, how many families would be expected to have (i) 3 boys and 2 girls (ii) 2 boys and 3 girls (iii) no girl (iv) at the most two girls.

[JNTU Nov. 2012 (Set No. 4)]

26. In a binomial distribution consisting of 5 independent trials, probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter 'p' of the distribution.

[JNTU (A) June 2018]

ANSWERS

1. 0.6243 2. 0.2461 3. $\frac{3}{16}$ 4. 0.9822 6. 6 7. 0.1937
8. (i) 0.3487 (ii) 0.3874 (iii) 0.9298 9. (i) 0.109375 (ii) 0.9648
10. (i) 0.890625 (ii) 0.65625 11. (i) 0.26 (ii) 0.0000244 (iii) 0.7674
12. (i) 0.2301 (ii) 0.3412 (iii) 0.2833
13. (i) 0.02642 (ii) 2, 0.0457 (iii) 1.023×10^{-7}
14. (i) $\frac{1}{5^{10}}$ (ii) 0.8926 (iii) 0.1074 (iv) 0.879128
15. (i) 0.1669 (ii) 0.9999 (iii) 0.999999
16. (i) 0.000004526 (ii) 0.00107374 (iii) $\frac{1}{5^{12}}$ (iv) 0.999999 (v) 0.422249
17. 0.65536 18. (i) 0.105 (ii) 0.00308 19. $\frac{1}{2^6}, \frac{6}{2^6}, \frac{15}{2^6}, \frac{20}{2^6}, \frac{15}{2^6}, \frac{6}{2^6}, \frac{1}{2^6}$
20.

x	0	1	2	3	4	5
$p(x)$	100	500	1000	1000	500	100

 ; $\mu = 2.5, \sigma = 1.118$
21. 33, 162, 316, 309, 151, 29 22. $80 [0.8725 + 0.2175]^{10}$
23. (i) $128 [0.517 + 0.483]^7$ (ii) 1, 7, 21, 35, 35, 21, 7, 1